

Maths-1 - Proving Techniques (Part 2) Work-along Sheet

Proofs are to mathematics what spelling is to poetry.

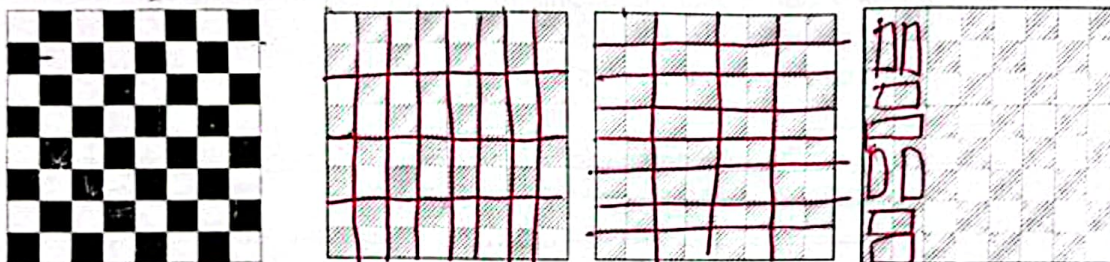
Mathematical works do consist of proofs, just as poems do consist of characters.

~ Vladimir Arnold

Name: Rishabh Baha ; Roll Number: _____
Lecture Batch: _____ ; Lab Batch: _____

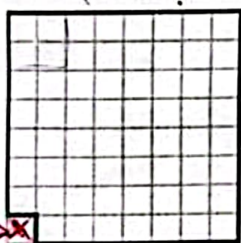
PROOF BY EXAMPLE:

Q. Tile the chessboard with 1×2 tiles.



MISSION ☒ POSSIBLE or MISSION IMPOSSIBLE?

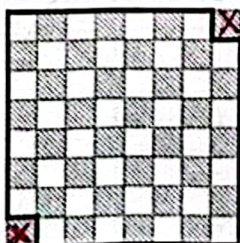
Q. Tile the chessboard without one of the corners with 1×2 tiles.



- There are 63 squares [31 black & 32 white]
- Each 1×2 tile covers one white square and one black square. So, each 1×2 tile covers 2 squares.
- 63 mod 2 = 1
- So, we will be unable to cover 1 square(s).

MISSION POSSIBLE or MISSION IMPOSSIBLE?

Q. Tile the following chessboard with 1×2 tiles:



62 squares $\begin{cases} \rightarrow 30B \\ \rightarrow 32W \end{cases}$

30 (1×2) tiles will cover 30B & 30W squares. ~~tiles~~
2 white squares are still pending.

MISSION POSSIBLE or MISSION IMPOSSIBLE?

Theorem:

"An 8x8 chessboard with two opposite corners removed cannot be tiled by 1x2 tiles".

Q. Can we tile the 8x8 board with two adjacent corners excluded (say, the bottom left square and top left square excluded from the board)?

Solution:

- There will be 31 black squares and 31 white squares.
- There are same/different number of black and white squares.
- So, we will be able/unable to tile the 8x8 board with two adjacent corners excluded.

Antecedent $P \rightarrow Q$ Conclusion Descendent

DIRECT PROOF: Assumption $\Leftarrow$ Hypothesis

A direct proof of a statement in the form " $p \rightarrow q$ " is constructed by assuming the truth of the hypothesis p and then logically deducing the truth of the conclusion q .

Q. Prove that if n is an odd integer, then n^2 is odd.

- $n = \underline{2k+1}$
- $n^2 = \underline{(2k+1)^2} = \underline{4k^2 + 4k + 1} = \underline{2(2k^2 + 2k) + 1}$
- Assume $\underline{2k^2 + 2k} = t$.
- So, we get $n^2 = \underline{2t+1}$ which is odd. Hence, proved!

Q. If m and n are both perfect squares, then $m \cdot n$ is also a perfect square.

- $m = \underline{k_1^2}$; $n = \underline{k_2^2}$
- $m \cdot n = \underline{k_1^2 \cdot k_2^2} = \underline{(k_1 \cdot k_2)^2}$ which indicates a perfect square.

> Implication : $p \rightarrow q$

> Converse : $q \rightarrow p$

> Inverse : $\neg p \rightarrow \neg q$

> Contrapositive : $\neg q \rightarrow \neg p$

Q. Check if the implication is equivalent to its contrapositive. same

p	q	$p \rightarrow q$	$\neg q$	$\neg p$	$\neg q \rightarrow \neg p$
T	T	T	F	F	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

From the above truth table, we can say that $p \rightarrow q \equiv \neg q \rightarrow \neg p$.

Q. Prove that if n is an integer and $3n + 2$ is odd, then n is odd.

➤ Can you try proving it using direct proof? [Yes / No]

➤ Is it time-consuming to use direct proof in this case? [Yes / No]

Try out using Proof by Contrapositive.

- $p = 3n+2$ is odd

- $q = \underline{n \text{ is odd}}$

- $p \rightarrow q$ $\equiv$ $\sim q \rightarrow \sim p$

- Alternative Equivalent Statement: If n is even, then $3n+2$ is even.

- $n = 2k$

$$\bullet \quad 3n + 2 = \underline{3(2k) + 2} = \underline{6k + 2} = \underline{2(3k + 1)}$$

- Assume $3k+1 = t$.

- So, we get $3n + 2 = \underline{2t}$ which is even.

- If the alternative equivalent statement is true, then the original statement is also true. Hence, we proved the given statement using the law of contrapositive

Importance of proper logic read from book

Q. Prove that if n is an integer and n^2 is odd, then n is odd.

- $p = \underline{n^2 \text{ is odd}}$

- $q = n$ is odd

- $p \rightarrow q$ $\equiv$ $\sim q \rightarrow \sim p$

- Alternative Equivalent Statement: If n is even, then n^2 is even.

• $n = 2k$

$$\bullet \quad n^2 = \frac{(2k)^2}{4} = \frac{4k^2}{4} = \frac{2(2k^2)}{4}$$

- Assume $2k^2$ = t.

• So, we get $n^2 = 2t$ which is even

- If the alternative equivalent statement is true, then the original statement is also true. Hence, we proved the given statement using the law of contrapositive.

★ Q. Prove that if n^2 is even, then n is even.

If n^2 is even, then n is even.

If n is odd, $\Downarrow$ then n^2 is odd.

$$n = 2k + 1$$

$$n^2 = (2k+1)^2 = 4k^2 + 4k + 1$$
$$= 2(2k^2 + 2k) + 1$$

$\underbrace{\hspace{10em}}_{\text{E (say)}}$

$$= \underline{\underline{2t+1}} \text{ which is odd.}$$

Hence, proved!

$$\sqrt{2} = \frac{a}{b} = \frac{2k_1}{2k_2} \quad a \text{ \& b are not co-prime.}$$

$$\gcd(a, b) \neq 1$$

So, our initial assumption was false.

Hence, $\sqrt{2}$ is irrational.

Q. Prove that $\sqrt{2}$ is irrational.

Assume $\sqrt{2}$ is rational.

$$\sqrt{2} = \frac{a}{b}, \quad a \in \mathbb{Z}, \quad b \in \mathbb{Z}, \quad b \neq 0$$

$$\frac{6}{8} = \left(\frac{3}{4}\right), \quad \frac{8}{16} = \left(\frac{1}{2}\right)$$

$\left(\frac{a}{b}\right)$ is in simplest form, i.e., $\gcd(a, b) = 1$

$$\sqrt{2} = \frac{a}{b} \quad \therefore 2 = \frac{a^2}{b^2} \quad \therefore a^2 = 2b^2$$

a^2 is even

$\downarrow$
a is even

a & b are co-prime, i.e., relatively prime.

$$a = 2k$$

$$\therefore (2k)^2 = 2b^2 \quad \therefore b^2 = 2k^2$$

$$\therefore 4k^2 = 2b^2$$

b^2 is even

$\downarrow$
b is even.

Q. Prove that there are infinitely many prime numbers.

Assume that there are finite prime numbers.

$p_1, p_2, p_3, \dots, p_n \leftarrow$ 'n' prime numbers

$$N = p_1 * p_2 * p_3 * \dots * p_n + 1$$

$$N \bmod p_1 = 1$$

$$N \bmod p_2 = 1$$

$$N \bmod p_3 = 1$$

$$\vdots$$

$$N \bmod p_n = 1$$

So, N is divisible by 1 & itself.

So, N is also a prime number.

So, we found $(n+1)^{\text{th}}$ prime number.

So, our initial assumption is false.
Hence, proved!

INDIRECT PROOF:

Indirect proofs utilise two main proving techniques: proof by contrapositive and proof by contradiction.

PROOF BY EXAMPLE (Revisited):

We use proof by example when the statement is of the form $\exists x P(x)$.

To prove such a statement, you only need to find one example c for which $P(c)$ is true.

Key point: Just one valid example is enough.

Q. Prove that there is an integer bigger than its cube.

$$P(x) = \underline{\hspace{2cm}}$$

Consider $x = \underline{\hspace{2cm}}$

$P(\underline{\hspace{2cm}})$: $\underline{\hspace{2cm}}$ which indicates $\underline{\hspace{2cm}}$ which is true.

Since we have found one example, the statement is proven true.